



FOUR-COURSE SYSTEM

## COURSE 3 - DATA ENGINEERING 2026

# C3-02 Targeted Remediation Cards

Repair exact weak competency -> changed micro-retest -> evidence

|              |                                                                                                          |
|--------------|----------------------------------------------------------------------------------------------------------|
| Artifact     | C3_02_TARGETED_REMEDIATION_CARDS_RC1                                                                     |
| Status       | 2026 - LEARNER EDITION                                                                                   |
| Phase        | C3-02 Transformation Engineering & Warehousing   14 lessons   34-46 focused hours                        |
| Prerequisite | C3-01 Gate PASS. C2B warehouse/modeling foundations may be assumed; implementation depth is taught here. |
| Brand        | Charcoal #1F2937   Teal #14B8A6   Soft Blue #3B82F6   AMOLED #000                                        |

House rule: analogy -> tiny example -> guided real tool -> break/fix -> independent changed version -> validation -> production reasoning -> evidence. A successful dbt command alone never proves correct data.



# T1 repair - Transformation & staging contracts

Layer boundaries, source/staging contracts and grains are explicit/replayable.

## REPAIR LOOP

1. Write the failed evidence/contract. 2. Reopen only the relevant C3-030 section. 3. Rebuild one tiny worked example. 4. Close all answers. 5. Do a changed micro-task in a different domain/key/data. 6. Run independent validation. 7. Save a new remediation attempt.

## PASS TO RETURN

Changed micro-task is independently correct, validation catches a deliberate plausible defect, explain-back is clear, and no critical failure remains.



## T2 repair - Deterministic models & DAG

Dependencies and current-state selection are deterministic and reviewable.

### REPAIR LOOP

1. Write the failed evidence/contract. 2. Reopen only the relevant C3-030 section. 3. Rebuild one tiny worked example. 4. Close all answers. 5. Do a changed micro-task in a different domain/key/data. 6. Run independent validation. 7. Save a new remediation attempt.

### PASS TO RETURN

Changed micro-task is independently correct, validation catches a deliberate plausible defect, explain-back is clear, and no critical failure remains.



## T3 repair - Testing & integrity

Grain/required/relationship/business tests block true defects; reconciliations prove no silent loss/fan-out.

### REPAIR LOOP

1. Write the failed evidence/contract. 2. Reopen only the relevant C3-030 section. 3. Rebuild one tiny worked example. 4. Close all answers. 5. Do a changed micro-task in a different domain/key/data. 6. Run independent validation. 7. Save a new remediation attempt.

### PASS TO RETURN

Changed micro-task is independently correct, validation catches a deliberate plausible defect, explain-back is clear, and no critical failure remains.



## T4 repair - Incremental & recovery

Update/insert + unchanged rerun safe; rebuild/full-refresh recovery documented.

### REPAIR LOOP

1. Write the failed evidence/contract. 2. Reopen only the relevant C3-030 section. 3. Rebuild one tiny worked example. 4. Close all answers. 5. Do a changed micro-task in a different domain/key/data. 6. Run independent validation. 7. Save a new remediation attempt.

### PASS TO RETURN

Changed micro-task is independently correct, validation catches a deliberate plausible defect, explain-back is clear, and no critical failure remains.



## T5 repair - History

Required descriptive change preserved as legitimate current+historical truth.

### REPAIR LOOP

1. Write the failed evidence/contract. 2. Reopen only the relevant C3-030 section. 3. Rebuild one tiny worked example. 4. Close all answers. 5. Do a changed micro-task in a different domain/key/data. 6. Run independent validation. 7. Save a new remediation attempt.

### PASS TO RETURN

Changed micro-task is independently correct, validation catches a deliberate plausible defect, explain-back is clear, and no critical failure remains.



## T6 repair - Dimensional/late-reference integrity

Facts/dimensions preserve declared grain; late references handled without silent fact loss.

### REPAIR LOOP

1. Write the failed evidence/contract. 2. Reopen only the relevant C3-030 section. 3. Rebuild one tiny worked example. 4. Close all answers. 5. Do a changed micro-task in a different domain/key/data. 6. Run independent validation. 7. Save a new remediation attempt.

### PASS TO RETURN

Changed micro-task is independently correct, validation catches a deliberate plausible defect, explain-back is clear, and no critical failure remains.



## T7 repair - Performance/CI/handoff

Plan reasoning, blocking CI/change safety and reproducible serving handoff are evidenced.

### REPAIR LOOP

1. Write the failed evidence/contract. 2. Reopen only the relevant C3-030 section. 3. Rebuild one tiny worked example. 4. Close all answers. 5. Do a changed micro-task in a different domain/key/data. 6. Run independent validation. 7. Save a new remediation attempt.

### PASS TO RETURN

Changed micro-task is independently correct, validation catches a deliberate plausible defect, explain-back is clear, and no critical failure remains.